Flink Data Collector

Cloud Insights

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Table of Contents

Flink Data Collector 1
- Installation 1
- Setup 3
- Objects and Counters 3
- Troubleshooting 8

Flink Data Collector

Cloud Insights uses this data collector to gather metrics from Flink.

Installation

1. From **Admin > Data Collectors**, click **+Data Collector**. Under **Services**, choose Flink.

   Select the Operating System or Platform on which the Telegraf agent is installed.

2. If you haven't already installed an Agent for collection, or you wish to install an Agent for a different Operating System or Platform, click *Show Instructions* to expand the Agent installation instructions.
3. Select the Agent Access Key for use with this data collector. You can add a new Agent Access Key by clicking the **+ Agent Access Key** button. Best practice: Use a different Agent Access Key only when you want to group data collectors, for example, by OS/Platform.
4. Follow the configuration steps to configure the data collector. The instructions vary depending on the type of Operating System or Platform you are using to collect data.

Flink

Flink Configuration

Gathers Flink metrics.

What Operating System or Platform Are You Using?

Need Help?

Windows

Select existing Agent Access Key or create a new one

Default (405fb5ec-d4cb-4404-977b-71fa931e1ad3)

+ Agent Access Key

*Please ensure that you have a Telegraf Agent in you environment before configuring Show Instructions

Follow Configuration Steps

Need Help?

1. Install Jolokia on your Flink JobManager(s) and Flink Task Manager(s). For details refer to the following document.

2. Copy the contents below into a new .conf file under the C:\Program Files\telegraf\telegraf.d\ folder. For example, copy the contents to the C:\Program Files\telegraf\telegraf.d\cloudinsights-flink.conf file.

```
    ## ##########################
    ## JobManager
    ## ##########################
    [[inputs.jolokia2_agent]]
      ## USER-ACTION: Provide address(es) of flink Job Manager(s), port for jolokia, add one URL
```

3. Replace <INSERT_FLINK_JOBMANAGER_ADDRESS> with the applicable Flink Job Manager address(es). Please specify a real machine address, and refrain from using a loopback address.

4. Replace <INSERT_FLINK_TASKMANAGER_ADDRESS> with the applicable Flink Task Manager address(es). Please specify a real machine address, and refrain from using a loopback address.

5. Replace <INSERT_JOLOKIA_PORT> with the applicable jolokia port.

6. Modify 'Namespace' if needed for server disambiguation (to avoid name clashes).

7. Modify 'Cluster' if needed for Flink cluster designation.

8. Restart the Telegraf service.

```
Stop-Service -Name telegraf -ErrorAction SilentlyContinue; Start-Service -Name telegraf
```

Setup

A full Flink deployment involves the following components:

JobManager: The Flink primary system. Coordinates a series of TaskManagers. In a High Availability setup, system will have more than one JobManager.
TaskManager: This is where Flink operators are executed.
The Flink plugin is based on the telegraf's Jolokia plugin. As such as a requirement to gather info from all Flink components, JMX needs to be configured and exposed via Jolokia on all components.

Compatibility

Configuration was developed against Flink version 1.7.0.

Setting Up

Jolokia Agent Jar

For all individual components, a version the Jolokia agent jar file must be downloaded. The version tested against was Jolokia agent 1.6.0.

Instructions below assume that downloaded jar file (jolokia-jvm-1.6.0-agent.jar) is placed under location '/opt/flink/lib/'.

JobManager

To configure JobManager to expose the Jolokia API, you can setup the following environment variable on your nodes then restart the JobManager:

```
export FLINK_ENV_JAVA_OPTS="-javaagent:/opt/flink/lib/jolokia-jvm-1.6.0-agent.jar=port=8778,host=0.0.0.0"
```

You can choose a different port for Jolokia (8778). If you have an internal IP to lock Jolokia onto you can replace the "catch all" 0.0.0.0 by your own IP. Notice this IP needs to be accessible from the telegraf plugin.

TaskManager

To configure TaskManager(s) to expose the Jolokia API, you can setup the following environment variable on your nodes then restart the TaskManager:

```
export FLINK_ENV_JAVA_OPTS="-javaagent:/opt/flink/lib/jolokia-jvm-1.6.0-agent.jar=port=8778,host=0.0.0.0"
```

You can choose a different port for Jolokia (8778). If you have an internal IP to lock Jolokia onto you can replace the "catch all" 0.0.0.0 by your own IP. Notice this IP needs to be accessible from the telegraf plugin.

Objects and Counters

The following objects and their counters are collected:

Object:	Identifiers:	Attributes:	Datapoints:
Flink Task Manager	Cluster Namespace Server	Node Name Task Manager ID Node IP	Network Available Memory Segments Network Total Memory Segments Garbage Collection PS MarkSweep Count Garbage Collection PS MarkSweep Time Garbage Collection PS Scavenge Count Garbage Collection PS Scavenge Time Heap Memory Committed Heap Memory Init Heap Memory Max Heap Memory Used Thread Count Daemon Thread Count Peak Thread Count Thread Count Total Started
Flink Job	Cluster Namespace server Job ID	Node Name Job Name Node IP Last Checkpoint External Path Restarting Time	Downtime Full Restarts Last Checkpoint Alignment Buffered Last Checkpoint Duration Last Checkpoint Size Number of Completed Checkpoints Number of Failed Checkpoints Number of in Progress Checkpoints Number of Checkpoints Uptime

Object:	Identifiers:	Attributes:	Datapoints:
Flink Job Manager	Cluster Namespace Server	Node Name Node IP	Garbage Collection PS MarkSweep Count Garbage Collection PS MarkSweep Time Garbage Collection PS Scavenge Count Garbage Collection PS Scavenge Time Heap Memory Committed Heap Memory Init Heap Memory Max Heap Memory Used Number Registered Task Managers Number Running Jobs Task Slots Available Task Slots Total Thread Count Daemon Thread Count Peak Thread Count Thread Count Total Started

Object:	Identifiers:	Attributes:	Datapoints:
Flink Task	Cluster Namespace Job ID Task ID	Server Node Name Job Name Sub Task Index Task Attempt ID Task Attempt Number Task Name Task Manager ID Node IP Current Input Watermark	Buffers In Pool Usage Buffers In Queue Length Buffers Out Pool Usage Buffers Out Queue Length Number Buffers In Local Number Bufffers In Local Per Second Count Number Buffers in Local Per Second Rate Number Buffers In Remote Number Buffers In Remote Per Second Count Number Buffers In Remote Per Second Rate Number Buffers Out Number Buffers Out Per Second Count Number Buffers Out Per Second Rate Number Bytes In Local Number Bytes In Local Per Second Count Number Bytes In Local Per Second Rate Number Bytes In Remote Number Bytes In Remote Per Second Count Number Bytes In Remote Per Second Rate Number Bytes Out Number Bytes Out Per Second Count Number Bytes Out Per Second Rate Number Records In Number Records In Per Second Count Number Records In Per Second Rate Number Records Out Number Records Out Per Second Count Number Records Out Per Second Rate

Object:	Identifiers:	Attributes:	Datapoints:
Flink Task Operator	Cluster Namespace Job ID Operator ID Task ID	Server Node Name Job Name Operator Name Sub Task Index Task Attempt ID Task Attempt Number Task Name Task Manager ID Node IP	Current Input Watermark Current Output Watermark Number Records In Number Records In Per Second Count Number Records In Per Second Rate Number Records Out Number Records Out Per Second Count Number Records Out Per Second Rate Number Late Records Dropped Assigned Partitions Bytes Consumed Rate Commit Latency Avg Commit Latency Max Commit Rate Commits Failed Commits Succeeded Connection Close Rate Connection Count Connection Creation Rate Count Fetch Latency Avg Fetch Latency Max Fetch Rate Fetch Size Avg Fetch Size Max Fetch Throttle Time Avg Fetch Throttle Time Max Heartbeat Rate Incoming Byte Rate IO Ratio IO Time Avg (ns) IO Wait Ratio IO Wait Time Avg (ns) Join Rate Join Time Avg Last Heartbeat Ago Network IO Rate Outgoing Byte Rate Records Consumed Rate Records Lag Max Records per Request Avg Request Rate Request Size Avg Request Size Max Response Rate Select Rate Sync Rate Sync Time Avg Heartbeat Response Time

Troubleshooting

Additional information may be found from the Support page.

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